

This is a remote-connected video sensing/surveillance product with state-of-the-art functionality such as 3-cameras for different angles of Viewing and wide-angle Viewing. This Product has wired connection Ports like 4-USB 3.0 Ports and 2-USB 2.0 Ports for future Purposes for connecting more cameras as well Wireless connectivity of Wi-Fi, Cellular, and BLE, where Wi-Fi and cellular connectivity will be used for streaming cameras Wirelessly to users mobile device.

## Tech Specifications:

- **Processor – Qualcomm QCS 6490 SOC**
- OS: Android 9.0
- System Components
  - **Telit 4G LTE Module (LE910C4-WWX)**
  - Wi-Fi + BLE
- Communication Protocol: MQTT
- POE Injector
- OV13B10 Camera Sensor



Rapidise has a developed Multi-Tenant Custom Cloud Video Management Software to replace the existing third-party Portal. The portal is a comprehensive video streaming platform that enables users to manage and distribute video content seamlessly. It provides a user-friendly interface for uploading, organizing, and sharing videos securely. The portal offers robust features for video playback, customization, and analytics, making it an ideal solution for businesses and individuals in various industries.

## Our Role:

- PRD Creation
- System Architecture
- UI Design
- Frontend Development
- Backend Development
- Admin/Partner/Custom er Portal
- Database
- DevOps Activity
- Testing and Deployment

## Technologies:

- Frontend: ReactJS
- Backend: NodeJS
- Web Server: Nginx
- Database: PostgreSQL and MongoDB
- Cloud Infrastructure: Amazon Web Services (AWS)
- Video Encoding: FFmpeg



Rapidise engaged in designing and developing a state-of-the-art access control system tailored for discerning multi-family property owners who want a perfect balance of contemporary design and high-tech functionality. It boasts the largest color touchscreen available in both cellular and ethernet models, along with external camera integration that offers multiple photo views of visitors and live streaming video at the gate.

## Our Role:

- PRD Creation
- System Architecture
- Hardware Development
- Firmware Development
- Mechanical ID Design
- EVT Prototype and Testing
- Testing and Deployment
- Certifications – FCC, PTCRB, UL, AT&T, Verizon, IP65, IK10
- Mass Production

## Technologies:

- iMX 8M Plus
- LoRaWan Gateway
- STM 32 MCU
- Wiegand Interface
- Cellular & Ethernet
- 11" display
- 3MP Camera
- 3 Axis IMU Sensor
- Microphone & Speaker
- IP65 - outdoor rated



The Smart Controller provides complete access control intelligence in one compact device, making it easier than ever to manage building security without the need for traditional, bulky access hardware. By leveraging existing electrified door systems the smart lock integrates seamlessly into any environment while providing users with enhanced functionality, remote control, and a streamlined experience. The device connects seamlessly to cloud-based platform, providing businesses with full control over their access systems with minimal setup.

## Our Role:

- PRD & System Architecture
- Embedded Hardware Development
- Embedded firmware Development
- EVT Prototype and Testing
- Mass Production
- Functional Testing

## Technologies:

- Authentication IC - SE050E2HQ1/Z01Z3Z
- IoT Module - WT02C40C
- WiFi-6 and BLE-5.4



The Project was developed to open and close the gate through LoRaWAN, BLE and HID reader cards. To achieve this, we built the hardware which can connect with LoRaWAN gateway as a Node, BLE peripheral which can connect with Mobile phone and HID wiegand interface to read the HID card data. The HID card data and other access code information can be stored in SD card. USB has been given to send debug messages on PC. Accelerometer is used to detect tampering event by detecting shake movement.

## Our Role:

- PRD Creation
- System Architecture
- Hardware Development
- Firmware Development
- PCB Fabrication, Component Procurement and PCB Assembly
- Board Bring Up
- Testing and Deployment

## Technologies:

- LoRaWAN Node
- BLE
- USB
- Wiegand Interface



SmarTack is a compact, Wi-Fi smart switch designed to be a gateway into the world of home automation. Tailored to fit seamlessly into standard 4M electric switch boxes, Smart Tack transforms traditional switches into intelligent, app- and voice-controlled devices. With robust integration into the Smart App, users can manage all their smart appliances, lighting, fans, air conditioners, geysers, and curtains, directly from their smartphones or via voice assistants like Siri, Alexa, or Google Assistant.

### Our Offerings

- System Architecture
- Embedded Hardware Development
- Embedded Firmware Development
- Cloud Architecture
- Mobile App(Android and iOS)
- Mechanical Enclosure
- Custom Web dashboards (Member, Admin, Users)

### Software Technologies

- iOS - Swift
- Android - Kotlin
- Backend - NodeJS
- Frontend - ReactJS
- Database - MongoDB
- Cloud - AWS
- Communication Protocol - MQTT, Wi-Fi
- Voice Assistant - Alexa, Google Home, Siri



Smart Touch, being a premium smart switch with Touch LCD, has a sleek shiny crystal look that adds a touch of distinction to your home, office, or enclosed space. It has a 10.1" crystal clear display providing sharp backgrounds and interactive touch controls. Our Google Home supported smart device takes it up a notch by allowing you to control your smart home devices with your voice. Simply ask Google Home to turn off the lights, adjust the temperature, or start your coffee maker. With Smart Touch with LCD, you can create a smart home that is truly customized to your needs and preferences.

### Our Offerings

- System Architecture
- Embedded Hardware Development
- Embedded Firmware Development
- Cloud Architecture
- Mobile App(Android and iOS)
- Mechanical Enclosure
- Custom Web dashboards (Member, Admin, Users)

### Software Technologies

- iOS - Swift
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- Backend - NodeJS
- Frontend - ReactJS
- Database - MongoDB
- Cloud - AWS
- Communication Protocol - MQTT, Wi-Fi
- Voice Assistant - Alexa, Google Home, Siri



Rapidise engaged in this project for Redesigning the existing device with new updated features with reduction of size, the device is used in power stations where main control unit is connected with this device for controlling the Lights, Gates and power connections. The device is simply used as relay controller and feedback system for the main controlling and processing unit.

## Our Role:

- PRD Creation
- System Architecture
- Hardware Development
- Firmware Development
- PCB Fabrication, Component Procurement and PCB Assembly
- Board Bring Up
- Testing and Deployment
- EMI/EMC Testing
- Reliability Testing
- CE, FCC Certification

## Technologies:

- Microchip Micro Controller
- BareMetal Firmware
- Ethernet Programming
- Relays
- UART Programming



Rapidise has developed a groundbreaking unified fall detection and vital capture system that integrates cutting-edge technologies to enhance the safety and well-being of veterans and elderly individuals. This case study provides an overview of the key functionalities and features of Fall Detection and Emergency Alert Response, Privacy-Preserving Video Clips, Real-time Vital Capture, Natural Language Understanding (NLU) Based Secondary Check, and Smart Lock Integration.

## Our Role:

- PRD & System Architecture
- Tejas Board 625 Carrier Development
- Firmware Development (AOSP)
- Device Side Application Development
- Mechanical Development
- AI Algorithm Development
- PCB Fabrication, Component procurement and PCB Assembly
- Testing and Deployment

## Technologies:

- Tejas 625 SOM
- Android 9.0
- Fall Detection Radar Module
- Communication Protocol: MQTT
- Far Field Mic Array
- USB to Ethernet Bridge IC



2.5D face recognition combines camera and Time-of-Flight sensor technology for enhanced facial recognition. By capturing depth information alongside traditional 2D imagery, it achieves superior accuracy and security. This enables facial authentication even in low-light conditions, making it ideal for secure access control and device unlocking. Its ability to distinguish between real faces and 2D images ensures robust protection against spoofing attempts, setting a new standard in biometric security.

## Key Features :

### 1. Model Architecture:

- a. Custom CNN model inspired from centernet architecture
- b. Detects keypoints and embeddings
- c. Light Weight for edge device | Less than 10 mb size

### 2. Platform: Qualcomm 8155

### 3. Accuracy Details:

- a. Accuracy: 96.2%
- b. Frames to Register new face: 2 frames
- c. Inference time: less than 1 sec



## Tech Stacks:

**Pre-Processing Libraries:** Opencv | NumPy | Pandas

**Seaborn Training Framework:** TensorFlow

**Model Optimization Tools:** TensorFlow Lite

**Runtime Framework:** TensflowLite | nvidia | Android

**Demo Video:** [Demo Video.mp4](#)

The Person Detection Sensor is a compact and cost-effective hardware module designed to detect nearby individuals' faces and provide essential information about their presence and location relative to the device. This case study focuses on the implementation of the Person Sensor's computer vision capabilities, primarily for facial tracking.

## Our Role:

- PRD Creation
- System Architecture
- Hardware Development
- Firmware Development
- PCB Fabrication, Component Procurement and PCB Assembly
- Board Bring Up
- Testing and Deployment

## Technologies:

- SSC335DE Sigmastar SOC,
- Camera Module
- Mic
- UART Programming



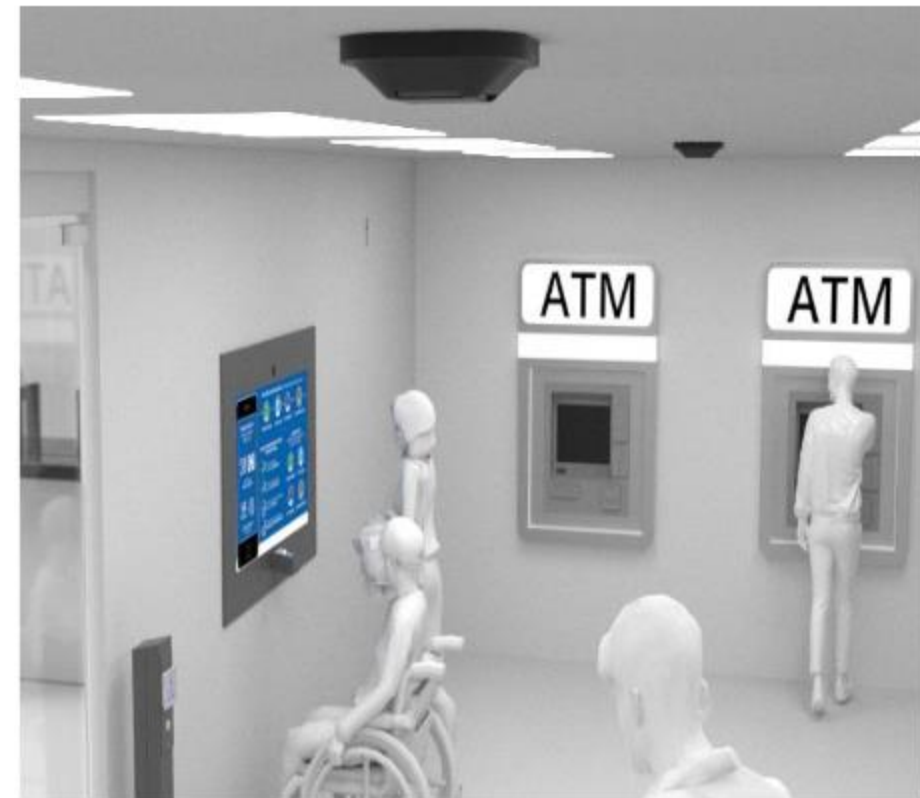
Rapidise engaged in developing a software for a RADAR-based Human Presence Detection (HPD) device for active monitoring of public spaces, ensuring social distancing and detecting loitering. The HPD is a high sensitive ceiling-mounted device used for monitoring activity in secure areas, such as ATM lobbies. The software detect and notify loiterers in ATM/tellerless branches, initiate loitering detection, and facilitate social distancing by limiting access to single individuals or small groups. Additionally, the HPD inhibits door access after entry to maintain single customer presence and alert to potentially dangerous situations on the client's property.

## Our Role:

- Desktop Application Development
- UI Designing
- Application Rigorous Testing

## Technologies:

- Frontend: ReactJS
- Backend: NodeJS
- .Net
- C#
- html/CSS



Rapidise has developed a Mailbox Receiver to receive a package from a courier drone (could be UPS, Amazon, a 3rd party drone courier delivering hot Chinese food or chilled prescription medication), and take it inside the 'Lockbox' housed within which keeps the package warm or cold based upon the contents.

## Our Role:

- Embedded Hardware Development
- Embedded Firmware Development
- Mechanical Development
- UI Design
- iOS Mobile App Development
- Web Interface Development
- Cloud Development
- REST APIs Development
- Managed Services
- Platform Version Upgrade
- Performance Optimization

## Technologies:

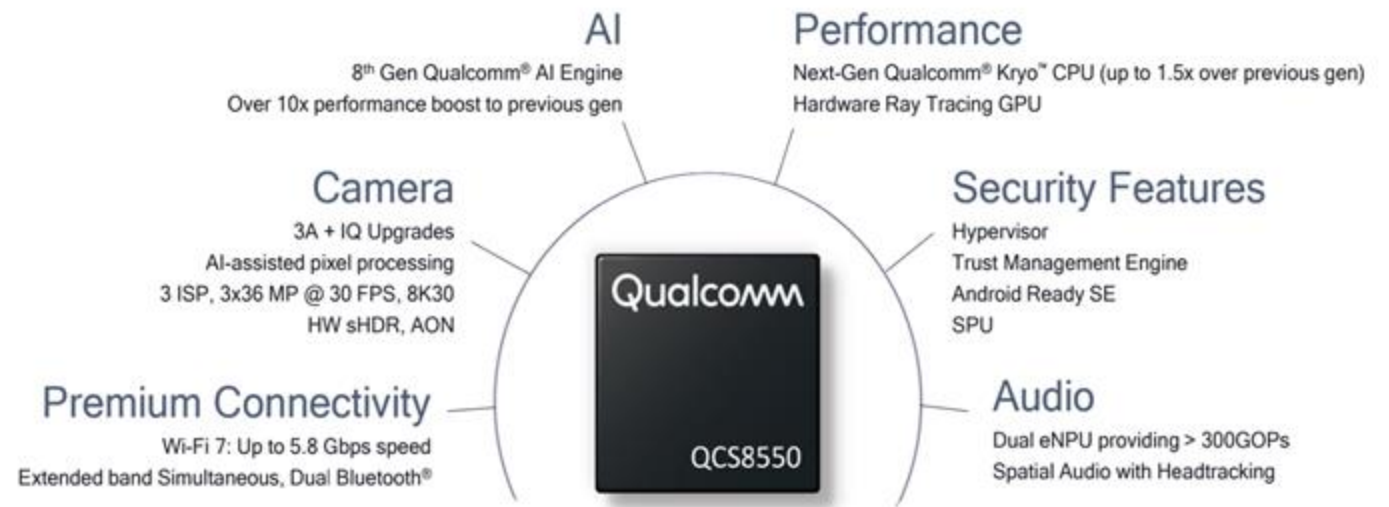
- Microprocessor
- GSM Implementation
- GPS Implementation
- Camera Integration
- Temperature sensor
- TOF Implementation
- Web Interface
- iOS - Swift
- Database - Hybrid (Mongo & MySQL)
- Cloud - AWS
- Cloud - Oracle
- Communication Protocol - MQTT



Rapidise is developing a Low Light Camera with new age Image Sensor with better Performance then current industrial offerings. The Project is to make a Portable , Modular form factor that can help in making this Product scale across multiple industries such as Defense Vehicles, Drones, night Vision Scopes, Underwater Photography and other mission critical use case. The main Value addition is that we are working on this is to translate the existing optimized pipeline to the Qualcomm advanced Spectra 7 ISP.

## Tech Specifications:

- Processor – **Qualcomm 8550 SOC**
- OS: Android 9.0
- System Components
  - LTE (4G) Communication
  - Wi-Fi 7
- QUAD Binning
- USB C display Port
- HDMI
- AI Algorithms
- 16 BIT 90 FPS



Next-generation premium SoC, delivering best-in-class system performance, exceptional intelligence, and premium imaging.

2.5D face recognition combines camera and Time-of-Flight sensor technology for enhanced facial recognition. By capturing depth information alongside traditional 2D imagery, it achieves superior accuracy and security. This enables facial authentication even in low-light conditions, making it ideal for secure access control and device unlocking. Its ability to distinguish between real faces and 2D images ensures robust protection against spoofing attempts, setting a new standard in biometric security.

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In this application, we aim to surveillance entity in their assigned working hours and also repost absence of his attributes, if any. As per the security compliance guidelines we make sure that every assigned entity is equipped with his assigned accessories such as helmet at his respective workplace. In case if a worker is not fulfilling his assigned protocols, then management will be notified and management can take right call to action in such scenarios.

### Our Role

- System Architecture
- Deep Learning
- Image Processing
- Embedded Hardware
- Embedded Firmware
- Cloud Architecture
- Database
- Android Application
- iOS Application

### Technologies

- Object Identification
- Deep Learning
- Tensorflow Lite
- CPM
- Computer Vision
- AWS

